Chapter 10

Time-Series Fundamentals

Dependence over time, stationarity, and the random walk

In this chapter. You will learn to describe a time series, to calculate the mean, autocovariance, and autocorrelation of a time-series model, to decide whether a model is stationary, and to read a sample autocorrelation function. Three models appear throughout: white noise, moving averages, and the random walk.

You will use. Covariance and correlation (Session 5), simulation and the central limit theorem (Session 6), and the idea of a stochastic process with a mean and a covariance function (Session 7).

Further reading. Pishro-Nik, Section 10.1.4, gives the formal definitions of stationarity.

How to use these notes. Read them before the exercises. The worked examples show every step of the calculations you will need. The short *Check* questions in the text test your understanding; answers are at the end of the chapter. Try each one before looking.

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10.1 What makes a time series special?

A **time series** is a sequence of measurements taken one after another at regular times:

$$x_1,\ x_2,\ \ldots,\ x_T.$$

Figure 10.1.1 shows four examples of the kind you meet in software engineering. Look at them for a moment before reading on.

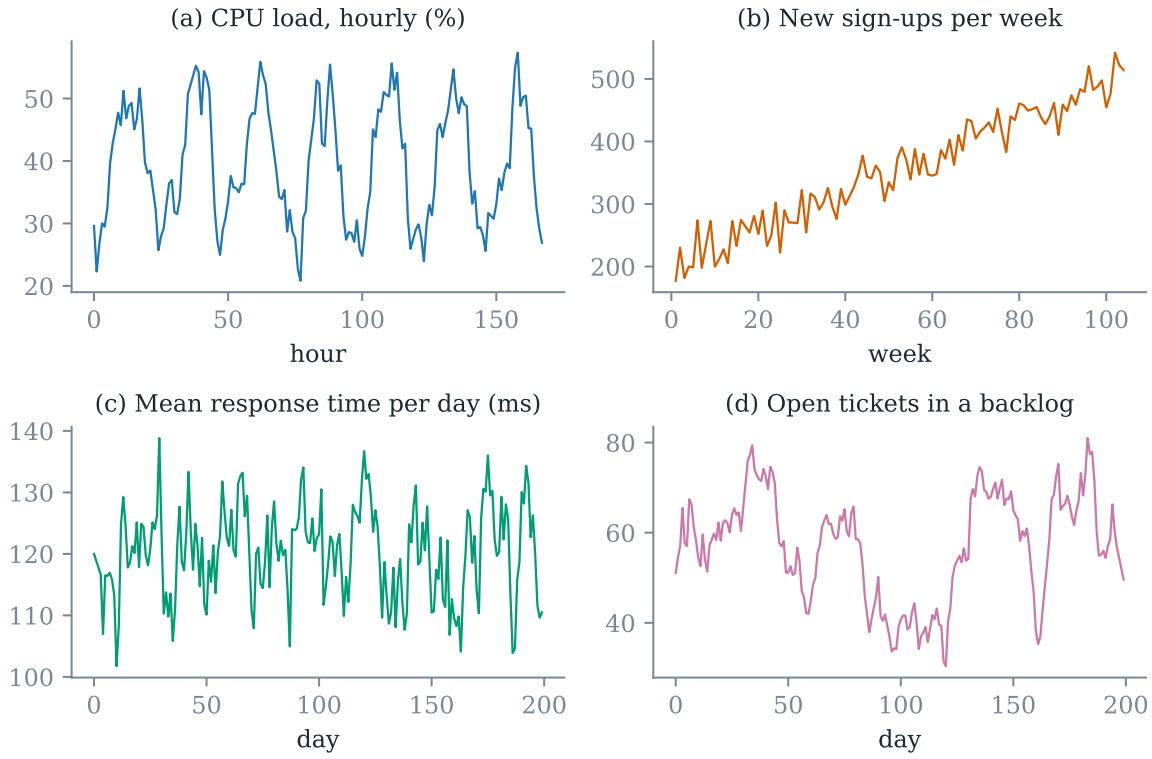


Figure 10.1.1. Four simulated time series. (a) Hourly CPU load over one week. (b) Weekly sign-ups over two years. (c) Daily mean response time. (d) The number of open tickets in a backlog.

What makes these data different from the samples of Session 6 is that the *order matters*. In a random sample, we can shuffle the observations without losing information. In a time series, shuffling destroys the most important feature: values close in time tend to be related. A busy hour on the server is usually followed by another busy hour; a long backlog today means a long backlog tomorrow.

Data and model

In Session 7 we called a family of random variables indexed by time a *stochastic process*. We now use this idea for time series.

- The **model** is a process $\{X_t\} = \{X_1, X_2, \ldots\}$. Each X_t is a random variable, and the variables have a joint distribution.
- The **data** $x_1, \ldots, x_T$ are *one realisation* of the process: the values that happened to occur.

We use capital letters for random variables and small letters for observed numbers, as in earlier sessions.

Key idea. We usually observe only *one* path of the process. We cannot rerun last week to see what else could have happened. Everything we learn from the data about dependence over time must come from this single path. Section 10.5 explains when that is possible.

10.2 Describing a series by eye

Always start by plotting the series against time. Three kinds of features are common.

Trend. A long-term rise or fall in the level, as in the sign-ups of Figure 10.1.1(b).

Seasonality. A pattern that repeats with a fixed, known period, such as the daily cycle in the CPU load of panel (a), with period 24 hours.

Irregular variation. What remains when trend and seasonality are removed. It is often still dependent over time, as in panel (c).

Panel (d) is different again: the backlog wanders without a fixed level and without a pattern. This is typical of a *random walk*, which we study in Section 10.9.

Check 1. For each series in Figure 10.1.1, decide whether it has a trend, a seasonal pattern, both, or neither. Which series look as if they fluctuate around the same level throughout?

10.2.1 Rolling mean and rolling variance

A simple tool to see whether the level or the spread changes over time is to compute the mean and the variance over a *moving window*, for example the last 30 values, and plot them against time.

In Figure 10.2.1, the rolling mean and variance of white noise wobble around constant values. For the random walk the rolling mean moves far from where it started, and the rolling variance changes a lot.

Watch out. Rolling statistics of a single path always wobble, even when the model behind them is perfectly stable, because each window contains only a few values. Ask whether the changes are larger than the wobble you would expect from a stable model. Simulating several paths from the model is a good way to find out.

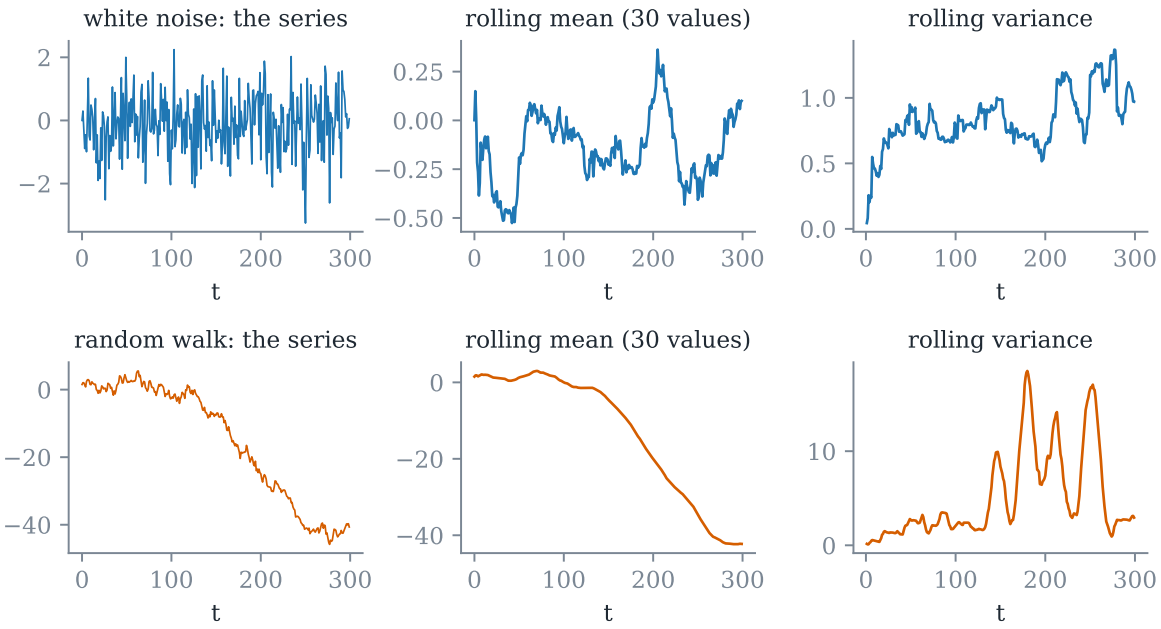


Figure 10.2.1. Rolling mean and rolling variance with a window of 30 values, for white noise (top) and a random walk (bottom).

10.3 Lag plots: does the past tell us about the present?

To see whether neighbouring values are related, plot each value against the one before it: the points (x_{t-1}, x_t). This is a **lag plot** at lag 1. More generally, the lag-k plot shows the points (x_{t-k}, x_t).

For white noise (left) the cloud is round: knowing yesterday's value tells you nothing about today's. For the moving average (middle) the cloud slopes upwards: high values tend to follow high values. At lag 2 (right) the relation is weaker.

The strength of a linear relation in a scatter plot is measured by a correlation coefficient. The correlation between X_t and X_{t-k} is called the **autocorrelation** at lag k: the correlation of the series with a copy of itself shifted k steps. The next section defines it for a model.

10.4 Mean, autocovariance, and autocorrelation

For a model $\{X_t\}$ we define, exactly as in Session 7,

$$\mu_t = \mathbb{E}[X_t] \qquad \text{and} \qquad \gamma(s,t) = \operatorname{Cov}(X_s, X_t).$$

The function γ is the **autocovariance function**. "Auto" indicates that both variables come from the same process.

Often the covariance depends only on how far apart the two times are, the **lag** $h = t - s$. Then we write it as a function of h alone.

Definition (autocovariance and autocorrelation). If $\mathbb{E}[X_t] = \mu$ for all t and $\operatorname{Cov}(X_t, X_{t+h})$

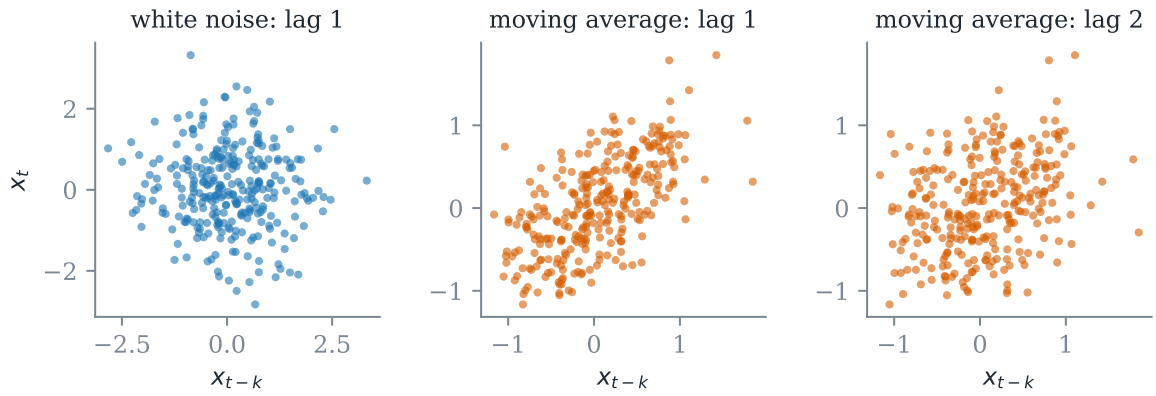


Figure 10.3.1. Lag plots for white noise and for a three-term moving average of white noise (defined in Section 10.7).

depends only on h, then

$$\gamma(h) = \operatorname{Cov}(X_t, X_{t+h}), \qquad \rho(h) = \frac{\gamma(h)}{\gamma(0)}. \tag{10.4.1}$$

γ is the autocovariance function and ρ the **autocorrelation function** (ACF) of the process.

The following properties hold for every such process. They follow from the rules for covariance in Session 5.

- $\gamma(0) = \operatorname{Cov}(X_t, X_t) = \operatorname{Var}(X_t)$, so $\rho(0) = 1$.
- $\gamma(-h) = \gamma(h)$: the covariance does not care which of the two variables comes first. We therefore only list $h \geq 0$.
- $-1 \leq \rho(h) \leq 1$, because $\rho(h)$ is a correlation.

Rules you will use again and again

Almost every calculation in this chapter uses only these rules for covariance, for constants a, b and random variables U, V, W:

$$\operatorname{Cov}(U, V) = \mathbb{E}[UV] - \mathbb{E}[U]\mathbb{E}[V], \tag{10.4.2}$$

$$\operatorname{Cov}(aU + bV,\ W) = a\operatorname{Cov}(U, W) + b\operatorname{Cov}(V, W), \tag{10.4.3}$$

$$\operatorname{Cov}(U + a,\ V + b) = \operatorname{Cov}(U, V), \tag{10.4.4}$$

$$\operatorname{Cov}(U, V) = 0 \quad \text{if } U \text{ and } V \text{ are independent.} \tag{10.4.5}$$

Adding a constant never changes a covariance; multiplying by constants multiplies the covariance.

Example 10.4.1 A trend plus noise

Let $\varepsilon_1, \varepsilon_2, \ldots$ be independent with mean 0 and variance σ^2, and let

$$X_t = 2 + 0.5t + \varepsilon_t.$$

Mean function. $\mathbb{E}[X_t] = 2 + 0.5t + \mathbb{E}[\varepsilon_t] = 2 + 0.5t$. It increases with t.

Autocovariance. The constant part $2 + 0.5t$ does not affect covariances, so $\text{Cov}(X_s, X_t) = \text{Cov}(\varepsilon_s, \varepsilon_t)$. This is σ^2 if $s = t$ and 0 otherwise.
So the covariances behave nicely, but the mean depends on time. We return to this example in Section 10.10.

10.5 Stationarity

A process is **stationary** if its probabilistic behaviour does not change over time. The time axis can be shifted without anyone noticing. There are two versions of this idea.

Definition (strict stationarity). $\{X_t\}$ is *strictly stationary* if for every number of time points k, every choice of times $t_1, \ldots, t_k$, and every shift h, the random vectors $(X_{t_1}, \ldots, X_{t_k})$ and $(X_{t_1+h}, \ldots, X_{t_k+h})$ have the same joint distribution.

Definition (weak stationarity). $\{X_t\}$ with finite variances is *weakly stationary* if

1. the mean is the same at all times: $\mathbb{E}[X_t] = \mu$, and
2. the covariance depends only on the lag: $\text{Cov}(X_t, X_{t+h}) = \gamma(h)$ for all t.

In particular the variance $\gamma(0)$ is the same at all times.

Strict stationarity concerns whole distributions; weak stationarity concerns only means and covariances, which are easier to check. In this course "stationary" means *weakly stationary* unless we say otherwise.

Key idea. To check weak stationarity: (1) calculate $\mathbb{E}[X_t]$ and see whether it depends on t; (2) calculate $\text{Cov}(X_t, X_{t+h})$ and see whether it depends on t or only on h. If t appears in either answer, the process is not stationary. One counterexample is enough.

10.5.1 Five examples

The examples below show that stationarity is not always what intuition suggests. In all of them, ε_t are independent with mean 0 and variance σ^2, and Z is a single random variable with mean 0 and variance 1.

Example 10.5.1 Constant level plus noise

Let $X_t = 4 + 2\varepsilon_t$. Then $\mathbb{E}[X_t] = 4$, $\text{Var}(X_t) = 4\sigma^2$, and for $h \neq 0$, $\text{Cov}(X_t, X_{t+h}) = 4\,\text{Cov}(\varepsilon_t, \varepsilon_{t+h}) = 0$. Nothing depends on t: the process is stationary.

Example 10.5.2 The same random number at every time

Let $X_t = Z$ for every t: we draw one number and repeat it forever (Figure 10.5.1a). Then $\mathbb{E}[X_t] = 0$ and $\text{Cov}(X_t, X_{t+h}) = \text{Var}(Z) = 1$ for every h. Neither depends on t, so the process is stationary, with $\rho(h) = 1$ for all h.

Each path is a flat line, but different realisations are at different heights. Stationary does *not* mean that the process forgets its past.

Example 10.5.3 An alternating sign

Let $X_t = (-1)^t Z$. The path jumps between Z and $-Z$ (Figure 10.5.1b). Then $\mathbb{E}[X_t] = (-1)^t \mathbb{E}[Z] = 0$ and

$$\operatorname{Cov}(X_t, X_{t+h}) = (-1)^t (-1)^{t+h} \operatorname{Var}(Z) = (-1)^{2t+h} = (-1)^h.$$

This depends only on h, so the process is stationary, with $\rho(h) = 1$ for even h and $\rho(h) = -1$ for odd h.

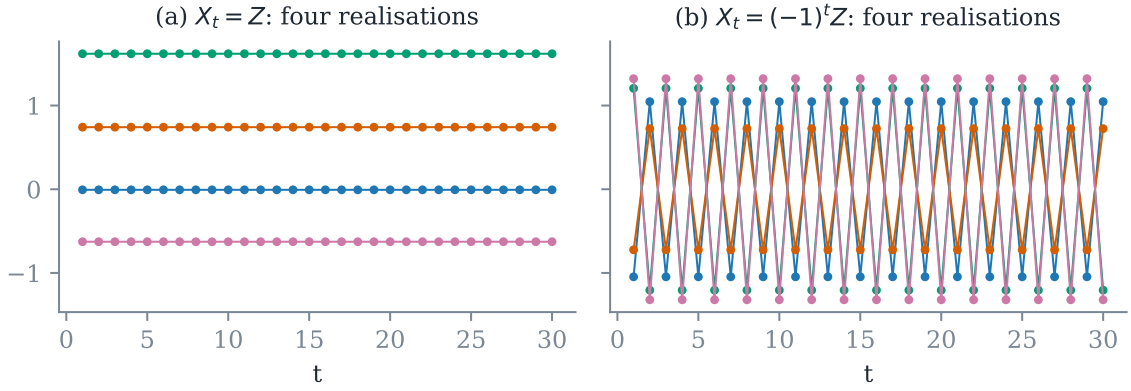


Figure 10.5.1. Four realisations each of $X_t = Z$ and of $X_t = (-1)^t Z$. Both processes are stationary.

Example 10.5.4 A growing variance

Let $X_t = t\,\varepsilon_t$. The mean is $\mathbb{E}[X_t] = 0$ for all t, but $\operatorname{Var}(X_t) = t^2\sigma^2$ grows with time. The process is not stationary. A constant mean alone is not enough.

Example 10.5.5 Weakly but not strictly stationary

Let the X_t be independent, with $X_t \sim N(0,1)$ for odd t and $X_t \sim \text{Uniform}(-\sqrt{3}, \sqrt{3})$ for even t. Both distributions have mean 0 and variance 1 (check: the uniform variance is $(2\sqrt{3})^2/12 = 1$), and different times are uncorrelated. So the process is weakly stationary. But it is not strictly stationary: the distribution of X_1 is normal and that of X_2 is uniform, so shifting time by one step changes the distribution.

Watch out. Stationary does not mean "flat", "constant", or "predictable". A stationary series can wander up and down, sometimes for a long time. What stays the same is the *probability law* that generates the values, not the values themselves.

Check 2. Is $X_t = t + \varepsilon_t - \varepsilon_{t-1}$ stationary? Is $Y_t = X_t - X_{t-1}$? (Use the rules from Section 10.4.)

10.5.2 Why stationarity matters

If a process is stationary, every stretch of the series is generated by the same mechanism. Then averaging over *time* along the one observed path tells us about the model, in the same way that averaging over *repetitions* did in Session 6. This is why the sample statistics of Section 10.11 make sense for stationary series, and why stationary models can be used for forecasting.

The same idea appeared for Markov chains in Session 9: in the stationary distribution, the probabilities of the states no longer change from step to step.

10.6 White noise

Definition (white noise). A process $\{\varepsilon_t\}$ is *white noise* with variance σ^2 if

$$\mathbb{E}[\varepsilon_t] = 0, \qquad \mathrm{Var}(\varepsilon_t) = \sigma^2, \qquad \mathrm{Cov}(\varepsilon_s, \varepsilon_t) = 0 \ \text{ for } s \neq t.$$

If the ε_t are moreover independent and identically distributed, we call it *i.i.d. noise*; if they are i.i.d. $N(0, \sigma^2)$, *Gaussian white noise*.

White noise is stationary, with

$$\gamma(0) = \sigma^2, \quad \gamma(h) = 0 \ \ (h \neq 0), \qquad \text{so} \qquad \rho(0) = 1, \quad \rho(h) = 0 \ \ (h \neq 0).$$

It is the time-series version of "pure randomness": past values carry no linear information about future values. The name comes from physics: like white light, it contains all frequencies equally.

White noise is the raw material for almost every model in this chapter and the next. We build dependent processes by combining white-noise terms.

Example 10.6.1 Uncorrelated but not independent

Let ε_t be i.i.d. $N(0,1)$ and define $W_t = \varepsilon_t \, \varepsilon_{t-1}$. Then $\mathbb{E}[W_t] = \mathbb{E}[\varepsilon_t]\mathbb{E}[\varepsilon_{t-1}] = 0$ and $\mathrm{Var}(W_t) = \mathbb{E}[\varepsilon_t^2]\mathbb{E}[\varepsilon_{t-1}^2] = 1$. For the lag-1 covariance,

$$\mathrm{Cov}(W_t, W_{t+1}) = \mathbb{E}[\varepsilon_{t-1} \, \varepsilon_t^2 \, \varepsilon_{t+1}] = \mathbb{E}[\varepsilon_{t-1}] \, \mathbb{E}[\varepsilon_t^2] \, \mathbb{E}[\varepsilon_{t+1}] = 0,$$

and longer lags are zero in the same way. So $\{W_t\}$ is white noise. But W_t and W_{t+1} are *not* independent: both contain ε_t, so $|W_t|$ and $|W_{t+1}|$ tend to be large together. White noise only requires zero correlation.

10.7 Building dependence: moving averages

A simple way to create a dependent process is to average neighbouring white-noise terms. The shared terms make neighbouring values similar.

Example 10.7.1 A three-term moving average

Let ε_t be white noise with variance σ^2, and

$$X_t = \tfrac{1}{3}\left(\varepsilon_t + \varepsilon_{t-1} + \varepsilon_{t-2}\right).$$

Step 1: mean. $\mathbb{E}[X_t] = \frac{1}{3}(0+0+0) = 0$.
Step 2: write out both variables. It helps to line up the coefficients of each white-noise term in a table:

	ε_{t+2}	ε_{t+1}	ε_t	ε_{t-1}	ε_{t-2}
X_t	0	0	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$
X_{t+1}	0	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	0
X_{t+2}	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	0	0

Step 3: multiply columnwise and add. Because different white-noise terms are uncorrelated, $\mathrm{Cov}(X_t, X_{t+h})$ is σ^2 times the sum over columns of the products of the two coefficients:

$$\begin{aligned}
\gamma(0) &= \sigma^2\left(\tfrac{1}{9} + \tfrac{1}{9} + \tfrac{1}{9}\right) = \tfrac{\sigma^2}{3}, \\
\gamma(1) &= \sigma^2\left(\tfrac{1}{3}\cdot\tfrac{1}{3} + \tfrac{1}{3}\cdot\tfrac{1}{3}\right) = \tfrac{2\sigma^2}{9} && \text{(columns } \varepsilon_t, \varepsilon_{t-1}\text{)}, \\
\gamma(2) &= \sigma^2\cdot\tfrac{1}{3}\cdot\tfrac{1}{3} = \tfrac{\sigma^2}{9} && \text{(column } \varepsilon_t\text{)},
\end{aligned}$$

and $\gamma(h) = 0$ for $|h| \geq 3$, since no column is shared.
Step 4: ACF. Divide by $\gamma(0)$:

$$\rho(1) = \tfrac{2/9}{1/3} = \tfrac{2}{3}, \qquad \rho(2) = \tfrac{1}{3}, \qquad \rho(h) = 0 \ \text{ for } |h| \geq 3.$$

None of these depends on t, so the process is stationary. In the simulated series of Figure 10.3.1 ($T = 300$) the sample autocorrelations were 0.63 and 0.30.

Example 10.7.2 A gap in the weights

Let $X_t = 5 + 2\varepsilon_t + \varepsilon_{t-2}$. The constant 5 is the mean and does not affect covariances. The coefficient table is

	ε_{t+2}	ε_{t+1}	ε_t	ε_{t-1}	ε_{t-2}
X_t	0	0	2	0	1
X_{t+1}	0	2	0	1	0
X_{t+2}	2	0	1	0	0

X_t and X_{t+1} have no column in which both weights are non-zero. X_t and X_{t+2} share only the column ε_t, with weights 2 and 1. Hence

$$\gamma(0) = (4+1)\sigma^2 = 5\sigma^2, \qquad \gamma(1) = 0, \qquad \gamma(2) = 2\sigma^2, \qquad \gamma(h) = 0 \ (|h| \geq 3),$$

so $\rho(1) = 0$ and $\rho(2) = 0.4$. A process can be uncorrelated at lag 1 and still correlated at lag 2.

Key idea. For a weighted sum of white-noise terms: line up the weights, keep only the terms that appear in *both* variables, and add (weight in X_t) $\times$ (weight in X_{t+h}) $\times \sigma^2$. A moving average has a *finite memory*: its ACF is exactly zero beyond the length of the window.

Check 3. Let $X_t = \varepsilon_t - \varepsilon_{t-1}$ with white-noise variance σ^2. Find $\gamma(0)$, $\gamma(1)$, and $\rho(1)$.

10.8 Same distribution, different dependence

The distribution of a single value X_t is called the **marginal distribution**. Two processes can have exactly the same marginal distribution and still behave completely differently over time.

Example 10.8.1 A "sticky" process

Process A consists of independent Uniform$(0,1)$ values. Process B starts with a Uniform$(0,1)$ value; at each step it keeps its current value with probability 0.9 and otherwise draws a fresh, independent Uniform$(0,1)$ value.

Marginal distribution. At every time, B is either an old uniform value or a new one, so $X_t \sim$ Uniform$(0,1)$ for both processes. The histograms of X_{40} over many realisations in Figure 10.8.1 are indistinguishable.

Dependence. For B, X_t and X_{t+h} are the same number if no fresh draw happened during the h steps, which has probability 0.9^h; otherwise they are independent. Using the law of total expectation (Session 5),

$$\operatorname{Cov}(X_t, X_{t+h}) = 0.9^h \cdot \operatorname{Var}(X_t) = \frac{0.9^h}{12}, \qquad \rho(h) = 0.9^h.$$

The simulation gives sample correlations 0.90 at lag 1 and 0.72 at lag 3, against 0.9 and $0.9^3 = 0.729$. Process A has $\rho(h) = 0$.

Key idea. A histogram pools the values and ignores their order, so it shows only the marginal distribution. Dependence over time is visible only in plots and statistics that use the order: the time plot, lag plots, and the ACF.

Process B is in fact a Markov process: the next value depends only on the current one. You will meet this structure again in Chapter 11.

10.9 The random walk

Instead of averaging white noise, we can *accumulate* it. Let $X_0 = 0$ and

$$X_t = X_{t-1} + \varepsilon_t, \qquad t = 1, 2, \ldots, \tag{10.9.1}$$

where ε_t is white noise with variance σ^2. Each step adds a new shock to everything that happened before. Unrolling the recursion,

$$X_t = \varepsilon_1 + \varepsilon_2 + \cdots + \varepsilon_t. \tag{10.9.2}$$

This is the sum S_t of Session 6, now regarded as a process in t.

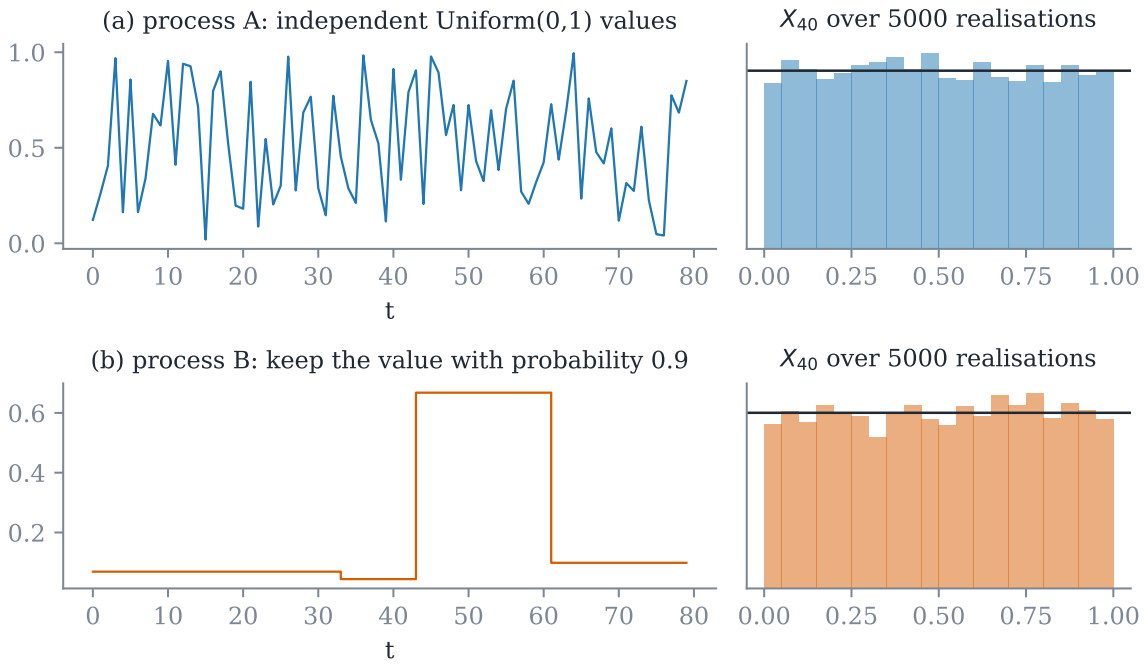


Figure 10.8.1. Two stationary processes with the same Uniform$(0,1)$ marginal distribution. Left: one path of each. Right: histograms of X_{40} over 5000 realisations.

Result (random walk). For a random walk with $X_0 = 0$ and $1 \le s \le t$,

$$\mathbb{E}[X_t] = 0, \qquad \mathrm{Var}(X_t) = t\sigma^2, \qquad \mathrm{Cov}(X_s, X_t) = s\sigma^2 = \sigma^2 \min(s,t). \tag{10.9.3}$$

Why. The mean and variance are those of a sum of t uncorrelated terms. For the covariance, split the later value into the earlier value plus the shocks in between:

$$X_t = X_s + (\varepsilon_{s+1} + \cdots + \varepsilon_t).$$

The bracket is uncorrelated with X_s, so $\mathrm{Cov}(X_s, X_t) = \mathrm{Cov}(X_s, X_s) = s\sigma^2$. This is the same argument as for the Poisson process in Session 7, where $\mathrm{Cov}(N(s), N(t)) = \lambda \min(s,t)$.

Example 10.9.1 How far does a random walk wander?

A backlog changes each day by a random amount with mean 0 and standard deviation $\sigma = 2$ tickets, independently of earlier days. Measure the backlog relative to today, so $X_0 = 0$.

After 25 days. $\mathrm{Var}(X_{25}) = 25 \cdot 4 = 100$, so the standard deviation is 10 tickets. If the daily changes are normal, $X_{25} \sim N(0, 10^2)$ and

$$\mathbb{P}(|X_{25}| > 10) = \mathbb{P}(|Z| > 1) \approx 0.317.$$

In a simulation of 20 000 paths the proportion was 0.320 (Figure 10.9.1, right).

Dependence between days. $\mathrm{Cov}(X_{16}, X_{25}) = 16 \cdot 4 = 64$ and

$$\mathrm{Corr}(X_{16}, X_{25}) = \frac{64}{\sqrt{64}\sqrt{100}} = 0.8.$$

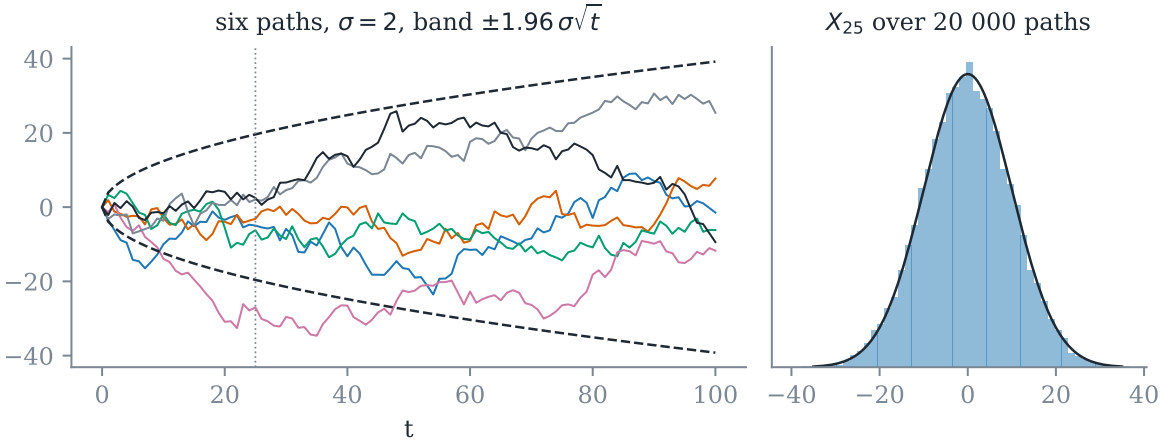


Figure 10.9.1. Left: six random-walk paths with $\sigma = 2$ and the band $\pm 1.96\sigma\sqrt{t}$. Right: the distribution of X_{25} over 20 000 paths, with the $N(0, 10^2)$ density.

The variance grows with t, and the covariance depends on s and t separately, not only on $t - s$. So **the random walk is not stationary**. Its paths do not return to a fixed level, and the band in Figure 10.9.1 widens like $\sqrt{t}$.

> **Key idea.** In general $\mathrm{Corr}(X_s, X_t) = s/\sqrt{st} = \sqrt{s/t}$. Two values of a long walk are strongly correlated even when they are far apart: after 100 days, days 100 and 110 have correlation $\sqrt{100/110} \approx 0.95$. A random walk never forgets.

Random walk with drift. If every step also adds a constant c, $X_t = c + X_{t-1} + \varepsilon_t$, then $X_t = ct + \varepsilon_1 + \cdots + \varepsilon_t$, so $\mathbb{E}[X_t] = ct$ and $\mathrm{Var}(X_t) = t\sigma^2$. With $c = 0.5$ and $\sigma = 2$, after 40 days $\mathbb{E}[X_{40}] = 20$ with standard deviation $2\sqrt{40} \approx 12.6$.

Connections. If the steps are $+1$ or -1, the random walk is a Markov chain on the integers (Session 8). In Chapter 11 it reappears as an autoregressive model with coefficient 1.

Check 4. For a random walk with $\sigma = 1$, compute $\mathrm{Var}(X_9)$, $\mathrm{Cov}(X_4, X_9)$, and $\mathrm{Corr}(X_4, X_9)$.

10.10 Differencing

Many non-stationary series become stationary if we look at their *changes* instead of their levels.

> **Definition (differences).** The *first difference* is $\nabla X_t = X_t - X_{t-1}$. For a seasonal period m (for example $m = 4$ for quarters or $m = 24$ for hours), the *seasonal difference* is $\nabla_m X_t = X_t - X_{t-m}$.

For data, differencing is just subtraction. The series 10, 12, 11, 15, 14 has first differences 2, -1, 4, -1.

Example 10.10.1 Differencing a random walk

For a random walk, $\nabla X_t = X_t - X_{t-1} = \varepsilon_t$: the differences are white noise. Conversely, if the differences of a series look like white noise, a random walk is a good model for the

series. Its best forecast is simply the last observed value.

Example 10.10.2 Differencing a trend

For $X_t = a + bt + \varepsilon_t$ (Example 10.4.1 had $a = 2$, $b = 0.5$),

$$\nabla X_t = \big(a + bt + \varepsilon_t\big) - \big(a + b(t-1) + \varepsilon_{t-1}\big) = b + \varepsilon_t - \varepsilon_{t-1}.$$

The trend has disappeared and the result is stationary. But it is not white noise: by Check 3, $\rho(1) = -\frac{1}{2}$. Differencing has *created* a negative autocorrelation that was not in the noise.

Example 10.10.3 Seasonal differencing

Let $X_t = s_t + \varepsilon_t$, where s_t repeats every four quarters ($s_t = s_{t-4}$). Then

$$\nabla_4 X_t = s_t - s_{t-4} + \varepsilon_t - \varepsilon_{t-4} = \varepsilon_t - \varepsilon_{t-4},$$

which is stationary, with $\rho(4) = -\frac{1}{2}$ and zero at all other lags $h \neq 0$. The pattern is removed completely.

In Figure 10.10.1, the differenced random walk has a sample ACF inside the band. The differenced trend has a clear spike at lag 1 ($r_1 = -0.47$), and the seasonally differenced series a spike at lag 4 ($r_4 = -0.41$), close to the theoretical -0.5.

Watch out. Difference only as much as necessary. Differencing a series that is already stationary introduces artificial dependence. Second differences, $\nabla^2 X_t = \nabla X_t - \nabla X_{t-1}$, are rarely needed.

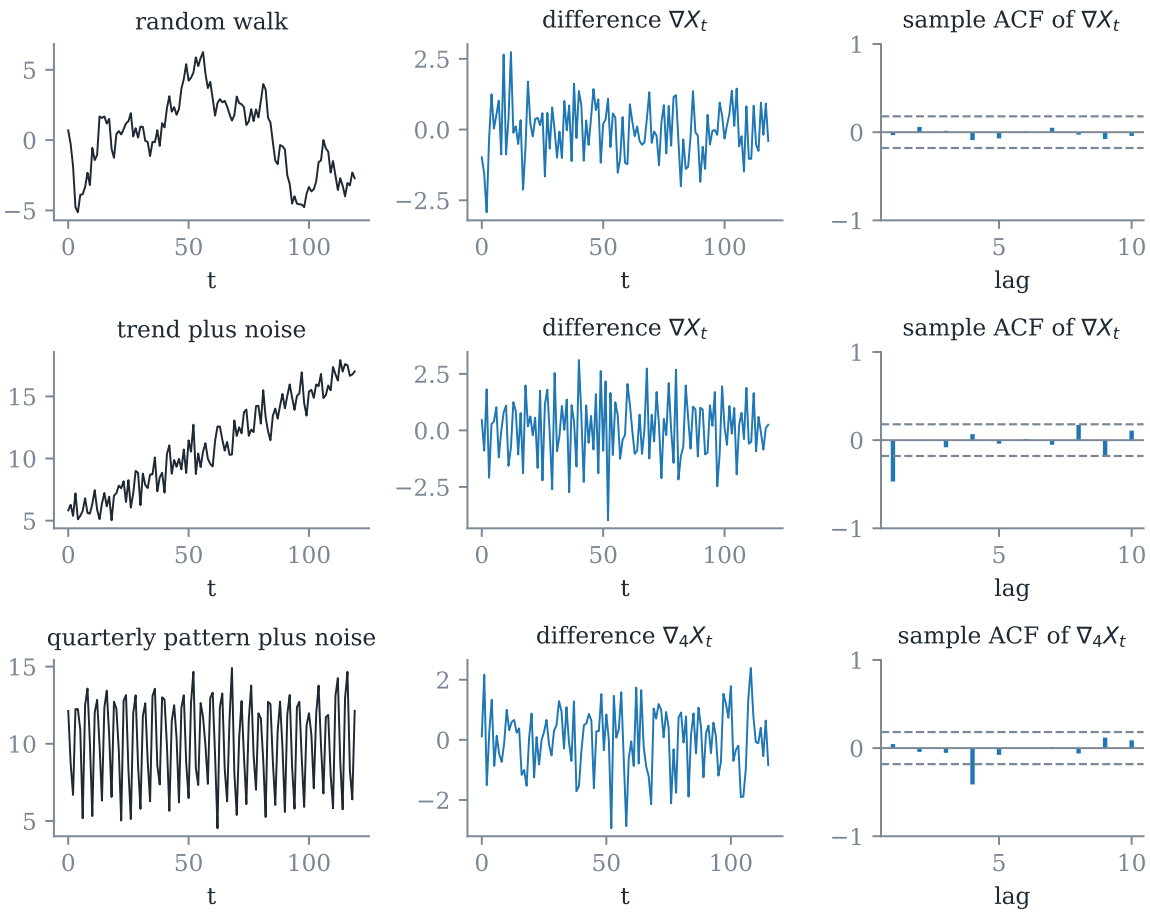


Figure 10.10.1. Three series, their differences, and the sample ACF of the differences. The dashed lines are the white-noise band of Section 10.12.

10.11 The sample ACF

So far $\gamma(h)$ and $\rho(h)$ were properties of a model. From observed data $x_1, \dots, x_T$ we compute their sample counterparts.

Definition (sample autocovariance and sample ACF). With $\bar{x} = \frac{1}{T}\sum_{t=1}^{T} x_t$,

$$c_k = \frac{1}{T}\sum_{t=k+1}^{T} (x_t - \bar{x})(x_{t-k} - \bar{x}), \qquad r_k = \frac{c_k}{c_0}. \tag{10.11.1}$$

The plot of r_k against k is the *sample ACF* or *correlogram*.

Notice three details:

- the *same* overall mean $\bar{x}$ is used in both factors;
- the sum at lag k has only $T - k$ terms;
- the denominator is T at every lag, not $T - k$.

This is the standard convention.

Example 10.11.1 A sample ACF by hand

Take $x = (1, 3, 4, 6, 7, 9)$, so $T = 6$ and $\bar{x} = 30/6 = 5$. Organise the calculation in a table of deviations $d_t = x_t - \bar{x}$:

t	1	2	3	4	5	6	sum
d_t	-4	-2	-1	1	2	4	
d_t^2	16	4	1	1	4	16	42
$d_t\, d_{t-1}$	–	8	2	-1	2	8	19
$d_t\, d_{t-2}$	–	–	4	-2	-2	4	4

Dividing each sum by $T = 6$:

$$c_0 = \tfrac{42}{6} = 7, \qquad c_1 = \tfrac{19}{6} \approx 3.17, \qquad c_2 = \tfrac{4}{6} \approx 0.67,$$

$$r_1 = \tfrac{19}{42} \approx 0.45, \qquad r_2 = \tfrac{4}{42} \approx 0.10.$$

The series rises steadily, so neighbouring deviations tend to have the same sign, and r_1 is positive.

Watch out. Different software uses different conventions, and the differences can be large for short series. For the data above, computing the ordinary (Pearson) correlation between the pairs (x_{t-1}, x_t), which uses separate means for the two sets of values, gives 0.97 instead of $r_1 = 0.45$. Always check which formula a function implements before comparing numbers.

The following function implements the formula of these notes:

```
import numpy as np

def sample_acf(x, max_lag):
    d = np.asarray(x, float) - np.mean(x)      # deviations from the overall mean
    denom = np.sum(d**2)                       # T * c_0
    return np.array([np.sum(d[k:] * d[:len(d)-k]) / denom
                     for k in range(max_lag + 1)])

sample_acf([1, 3, 4, 6, 7, 9], 2)              # array([1.    , 0.4524, 0.0952])
```

Check 5. Compute r_1 by hand for the series $x = (2, 0, 2, 0, 2, 0)$. What does the sign tell you?

10.12 How large is "large"? The white-noise band

Even if a series is white noise, its sample autocorrelations are not exactly zero. They are computed from random data, so they are random variables themselves, just like the sample mean in Session 6.

Result (sampling variation of r_k for white noise). If the data come from i.i.d. noise and T is large, each r_k with $k \geq 1$ is approximately $N(0, 1/T)$. Hence about 95% of the sample

autocorrelations lie within

$$\pm\frac{1.96}{\sqrt{T}}. \tag{10.12.1}$$

These are the dashed lines in the ACF plots of these notes. For $T = 50$ the band is ± 0.277; for $T = 200$ it is ± 0.139. Longer series give narrower bands, as the $1/\sqrt{n}$ rule of Session 6 predicts.

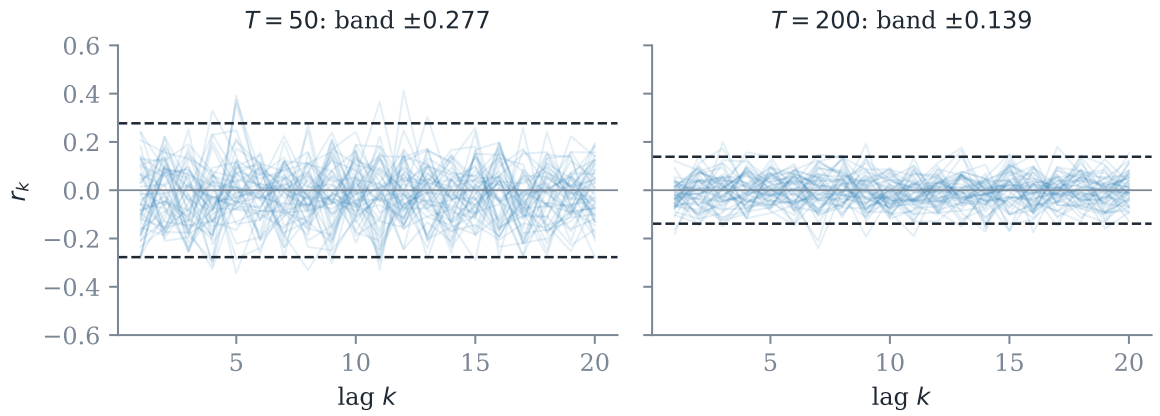


Figure 10.12.1. Sample ACFs of 50 simulated white-noise series, for two series lengths. All of these series have a true ACF of zero.

Example 10.12.1 What the band means in practice

In a simulation of 1000 white-noise series of length 200, the standard deviation of r_1 was 0.070, close to $1/\sqrt{200} = 0.071$. About 4% of all sample autocorrelations fell outside the band. But in 56% of the series, *at least one* of the first 20 lags fell outside.

So when you inspect 20 lags, one spike just outside the band is no evidence of dependence. A spike far outside the band, several spikes, or a systematic pattern (a slow decay, or peaks at the seasonal lags) is.

Watch out. The band is derived *assuming* the series is white noise. It is not a confidence band for the true $\rho(k)$ of other processes, and it is meaningless for non-stationary series. Formal tests based on these ideas belong to statistics and are not part of this course.

10.13 Reading a sample ACF

Figure 10.13.1 shows four series and their sample ACFs. Try to match each ACF to a type of process before reading the explanation.

Example 10.13.1 Diagnosing four series

A. A single clear spike at lag 2 ($r_2 = 0.35$) and nothing else: finite memory, as for a moving average. It is the process of Example 10.7.2, with $\rho(2) = 0.4$.

B. Large positive values that decay slowly: the signature of a trend.

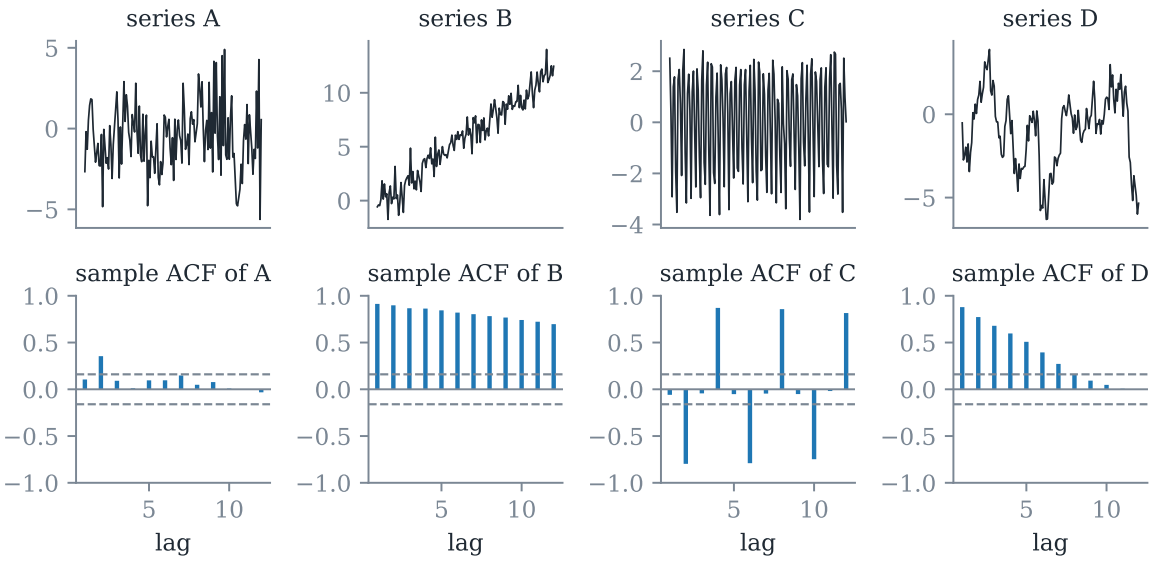


Figure 10.13.1. Four simulated series of length 150 and their sample ACFs.

C. Peaks at lags 4, 8, and 12, and negative values in between: a seasonal pattern with period 4.
D. Values close to 1 that decay very slowly: a random walk. The strong correlation between distant values comes from the shared accumulated shocks (Section 10.9).

Process	**Typical sample ACF**
White noise	All spikes small, mostly inside the band
Moving average over $q+1$ terms	Spikes up to lag q, then small
Trend	Large positive values, slow decay
Seasonality with period m	Peaks at lags $m, 2m, \ldots$
Random walk	Values near 1, very slow decay

Watch out. For a non-stationary series, the sample ACF still produces numbers, but there is no single $\rho(h)$ for them to estimate. A slowly decaying sample ACF is a warning sign: difference the series or model the trend first.

10.14 A stationary process that looks deterministic

Each path in Figure 10.14.1 looks like a smooth wave. Yet the process is stationary. This example also prepares calculations with a random phase.

Example 10.14.1 Warm-up: averages over a uniform angle

Let $U \sim \text{Uniform}(0, 2\pi)$, with density $1/(2\pi)$ on $[0, 2\pi]$. Then

$$\mathbb{E}[\cos U] = \frac{1}{2\pi}\int_0^{2\pi} \cos u \, du = \frac{1}{2\pi}\big[\sin u\big]_0^{2\pi} = 0,$$

and, using $\cos^2 u = \frac{1}{2}(1 + \cos 2u)$,

$$\mathbb{E}[\cos^2 U] = \frac{1}{2\pi}\int_0^{2\pi} \tfrac{1}{2}(1 + \cos 2u)\, du = \tfrac{1}{2}.$$

An average of a cosine over a whole period is zero.

Example 10.14.2 Random amplitudes

Let A and B be uncorrelated with mean 0 and variance σ^2, let ω be a fixed frequency, and

$$X_t = A\cos(\omega t) + B\sin(\omega t).$$

Mean. $\mathbb{E}[X_t] = \cos(\omega t)\mathbb{E}[A] + \sin(\omega t)\mathbb{E}[B] = 0$.
Covariance. Since $\text{Cov}(A, B) = 0$,

$$\text{Cov}(X_s, X_t) = \sigma^2\big(\cos\omega s\cos\omega t + \sin\omega s\sin\omega t\big) = \sigma^2\cos\big(\omega(t-s)\big),$$

by the identity $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$. It depends only on $t - s$: the process is stationary, with $\rho(h) = \cos(\omega h)$. A simulation with $\omega = 2\pi/12$ gave $\text{Cov}(X_3, X_5) \approx 0.504$, against $\cos(2\omega) = 0.5$.

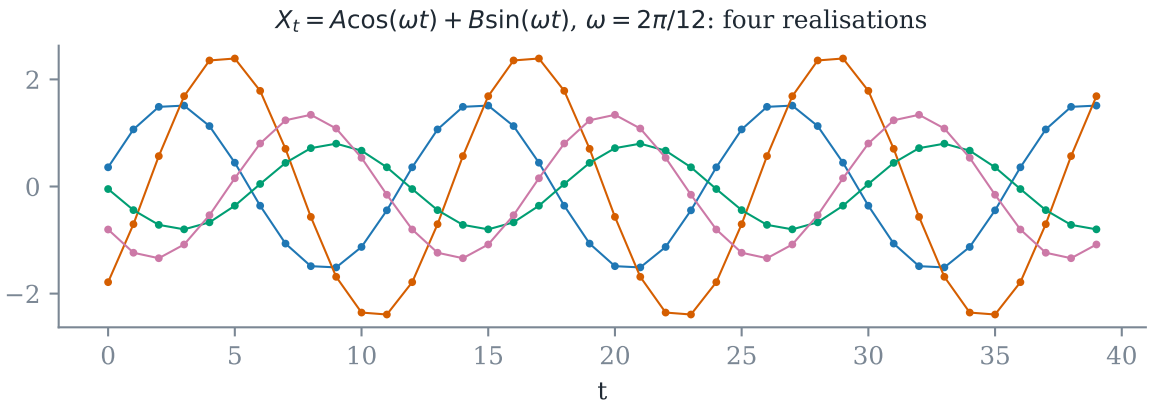


Figure 10.14.1. Four realisations of a stationary process with random amplitudes.

Key idea. Each path is a wave, but the amplitude and the position of the peaks are random. Averaged over all realisations, no time point is special. This is what stationarity means: the *law* is the same at all times, even if every path has a clear shape.

10.15 Simulating and checking in Python

Simulation is the best way to see the difference between a model and one realisation of it. The code below simulates white noise, a moving average, and a random walk, and compares their sample ACFs with the theory.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

rng = np.random.default_rng(1)
T, sigma = 300, 1.0
eps = rng.normal(0, sigma, T + 2)

wn = eps[2:]                                  # white noise
ma = (eps[2:] + eps[1:-1] + eps[:-2]) / 3     # Example 10.7.1
rw = np.cumsum(eps[2:])                       # random walk

print(sample_acf(ma, 4))                      # theory: 1, 2/3, 1/3, 0, 0

s = pd.Series(rw)
fig, ax = plt.subplots(3, 1, sharex=True)
ax[0].plot(s);                    ax[0].set_title("random walk")
ax[1].plot(s.rolling(30).mean()); ax[1].set_title("rolling mean")
ax[2].plot(s.rolling(30).var());  ax[2].set_title("rolling variance")
plt.show()
```

Run the simulation several times with different seeds. Features that appear every time are properties of the model; features that change from run to run are sampling variation.

10.16 Summary

Concept	What to remember
Time series	One realisation $x_1, \ldots, x_T$ of a process $\{X_t\}$
ACF of a model	$\gamma(h) = \mathrm{Cov}(X_t, X_{t+h})$, $\rho(h) = \gamma(h)/\gamma(0)$
Weak stationarity	Constant mean, and covariance depending only on the lag
White noise	Mean 0, variance σ^2, uncorrelated: $\rho(h) = 0$ for $h \neq 0$
Moving average	Line up weights, use shared terms; ACF zero beyond the window
Random walk	$\mathrm{Var}(X_t) = t\sigma^2$, $\mathrm{Cov}(X_s, X_t) = \sigma^2 \min(s,t)$; not stationary
Differencing	$\nabla X_t = X_t - X_{t-1}$, $\nabla_m X_t = X_t - X_{t-m}$
Sample ACF	$r_k = c_k/c_0$, denominator T; white-noise band $\pm 1.96/\sqrt{T}$

Looking ahead. A moving average forgets after a few steps; a random walk never forgets. Chapter 11 introduces the autoregressive model, whose memory fades gradually. It is stationary, and it is the basis for forecasting.

Answers to the checks

Check 1. (a) Daily seasonality with period 24, no clear trend. (b) A rising trend, no visible seasonality. (c) Neither: it fluctuates around a stable level, with some persistence. (d) No fixed level and no pattern: it wanders. Only (c) looks as if it has the same level throughout.

Check 2. $\mathbb{E}[X_t] = t$, so X_t is not stationary. For Y_t: $Y_t = 1 + \varepsilon_t - 2\varepsilon_{t-1} + \varepsilon_{t-2}$, with constant mean 1 and covariances that depend only on the lag (weights $1, -2, 1$), so Y_t is stationary.

Check 3. Weights 1 and -1: $\gamma(0) = 2\sigma^2$, $\gamma(1) = 1 \cdot (-1) \cdot \sigma^2 = -\sigma^2$, $\rho(1) = -\frac{1}{2}$.

Check 4. $\mathrm{Var}(X_9) = 9$, $\mathrm{Cov}(X_4, X_9) = 4$, $\mathrm{Corr}(X_4, X_9) = 4/(2 \cdot 3) = \frac{2}{3} = \sqrt{4/9}$.

Check 5. $\bar{x} = 1$, deviations $1, -1, 1, -1, 1, -1$, $\sum d_t^2 = 6$, $\sum d_t d_{t-1} = -5$, so $r_1 = -5/6 \approx -0.83$. The negative sign shows that high and low values alternate.